

Master Skills Taxonomy & Many-to-Many Mapping Specification

This document serves as the formal architectural specification for organizing, managing, and querying candidate technical skills within the University of Ruhuna's Department of Computer Science **CV Management and Internship Filtering System**.

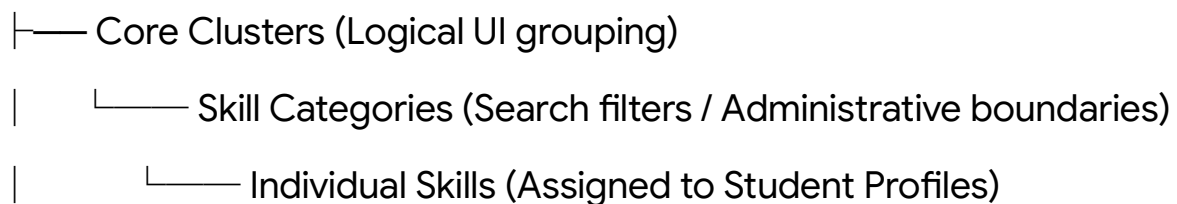
To ensure high data integrity and a seamless user experience, this design implements a **Many-to-Many Relationship model (Option A)**. This allows multi-purpose skills (such as Python, Java, or JavaScript) to reside as a single unique entry in the database while dynamically mapping to multiple operational categories.

1. System Taxonomy Architecture Overview

The taxonomy is organized across **5 High-Level Clusters**, which contain **18 Core Categories**, further mapping down to individual low-level skills.

Legend: Skills marked with an asterisk (*) represent cross-mapped skills that exist once in the core database but are linked to multiple categories via the SkillCategoryMapping entity.

[Master Skill Directory]



2. Master Cluster & Category Breakdown

Cluster 1: Core Engineering

1. Frontend Development

- **Description:** Client-side technologies, libraries, and frameworks used to construct responsive, highly interactive web applications.

- **Low-Level Skills:** React, Angular, Next.js, Vue, HTML5, CSS3, Tailwind CSS, Bootstrap, JavaScript*, TypeScript*.

2. Backend Development

- **Description:** Server-side environments, core programming languages, and robust APIs supporting application logic, processing, and system integrations.
- **Low-Level Skills:** Java*, Spring, Spring Boot, .NET, .NET Core, C#, Node.js*, Express.js, Python*, C++, Go, Ballerina, Ruby on Rails, PHP, JavaScript*, TypeScript*.

3. Mobile Development

- **Description:** Cross-platform and native environments used to deliver functional mobile operating system applications.
- **Low-Level Skills:** Flutter, React Native, Android Studio, Kotlin, Swift, iOS SDK, Java*.

4. Software Architecture & Design Patterns

- **Description:** High-level structural concepts and paradigms managing distributed, scalable, and modular software communications.
- **Low-Level Skills:** Microservices, REST APIs, Distributed Systems, Cloud Native, SOAP, GraphQL, Event-Driven Architecture, gRPC, MVC Architecture, Message Broker Integration*.

Cluster 2: Data & Infrastructure

5. Databases & Data Engineering

- **Description:** Relational and non-relational database management systems, caching layouts, and asynchronous data stream brokers.

- **Low-Level Skills:** Oracle, SQL Server / MSSQL, MySQL, PostgreSQL, MongoDB, Redis, Cassandra, Neo4j, SQL, NoSQL, Big Data, Kafka*, RabbitMQ*.

6. Cloud & DevOps

- **Description:** Infrastructure deployment, automation setups, continuous deployment pipelines, and cloud platform environments.
- **Low-Level Skills:** AWS, Azure, GCP, Docker, Kubernetes, Azure DevOps, GitHub Actions, Jenkins, CI/CD Pipelines, Terraform, Linux Shell Scripting*.

7. AI & Data Science

- **Description:** Computational libraries, neural pipelines, and high-level descriptive modeling tools.
- **Low-Level Skills:** Python*, R, Machine Learning, Deep Learning, Natural Language Processing (NLP), Data Analytics, Pandas, NumPy, TensorFlow, PyTorch, Matplotlib, Tableau.

Cluster 3: Product & Quality Assurance

8. UI/UX Design

- **Description:** Layout prototyping, wireframing, and interactive digital interfaces aligned with accessibility guidelines.
- **Low-Level Skills:** Figma, Adobe XD, Sketch, Wireframing, User Personas, Interaction Design, Prototyping, Heuristic Evaluation.

9. QA & Testing

- **Description:** Validations representing automated scripts, programmatic testing frameworks, and verification processes.

- **Low-Level Skills:** Manual Testing, QA Automation, Selenium, Postman, JUnit, Cypress, Bug Tracking, Integration Testing, Regression Testing, Unit Testing.

10. Business Analysis & Project Management

- **Description:** Operational frameworks managing requirements gathering, process modeling, and delivery schedules.
- **Low-Level Skills:** Agile/Scrum, Software Development Life Cycle (SDLC), Requirement Gathering, UML Modeling, User Stories, Jira, Confluence, Miro.

Cluster 4: Academic & Foundations

11. Computer Science Fundamentals

- **Description:** Core academic modules and computer science principles providing standard computational logic.
- **Low-Level Skills:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Database Management Systems (DBMS) Theory, Automata Theory, Compiler Design, Systems Design.

12. Systems, Networking & Cybersecurity

- **Description:** Low-level administration, infrastructure configurations, vulnerability assessments, and standard threat models.
- **Low-Level Skills:** Linux System Administration, Bash/Shell Scripting*, Cisco/CCNA, Network Security, Information Security, OWASP Top 10, Penetration Testing, Firewall Configuration.

13. Version Control & Collaboration

- **Description:** Distributed systems managing tracking, repository backups, and branch alignment workflows.

- **Low-Level Skills:** Git, GitHub, GitLab, Bitbucket.

14. Academic Research & Technical Writing

- **Description:** Scientific compilation, technical publishing, and advanced research documentation methods.
- **Low-Level Skills:** LaTeX, Technical Paper Writing, Research Methodologies, Statistical Data Analysis, Academic Presentation.

Cluster 5: Professional Attributes & Niches

15. Soft Skills & Core Competencies

- **Description:** Cognitive and interpersonal traits required for sustainable professional team collaboration.
- **Low-Level Skills:** Communication, Problem-Solving, Teamwork, Critical Thinking, Time Management, Adaptability, Leadership.

16. Certifications & Credentials

- **Description:** Industry-recognized credentials proving external validation of technical competencies.
- **Low-Level Skills:** AWS Certified Cloud Practitioner, Microsoft Certified Azure Fundamentals, Oracle Certified Associate (OCA), ISTQB Certified Tester.

17. Spoken & Written Languages

- **Description:** Language proficiencies facilitating interactions across domestic and international development pipelines.
- **Low-Level Skills:** English (Professional Proficiency), Sinhalese, Tamil.

18. Domain Knowledge & Industry Verticals

- **Description:** Contextual knowledge representing specific industries and niche product ecosystems.

- **Low-Level Skills:** Digital Banking Systems, Travel Software Systems, Stock Trading Platforms, Healthcare Systems, API Management, Identity Platforms, FinTech, E-Commerce.